

Homework 1

Due Sep 15, start of class

Write down your solutions and submit them as a single PDF to loss0f0706@gmail.com.
 Problem 6 is an optional bonus (20 points).

Problem 1. Reading and writing the language**(20 points)**

A transport aircraft is being loaded for a resupply flight. Three cargo types:

	water pallet	fuel drum	medical pallet
value (pts)	5	4	7
weight (t)	2	3	1
volume (m ³)	3	1	2

The aircraft carries at most 60 t and 55 m³. Orders require that *exactly* 4 medical pallets fly. For this problem cargo is divisible — fractional pallets may be manifested. The goal is to maximize total value.

- (6 pts)** Define the decision variables and write the problem as a scalar LP: objective plus constraints, one line each, with a few words saying what each constraint enforces.
- (8 pts)** Put your LP in the standard form $\max c^\top x$ s.t. $Ax \leq b$ from Week 1. Write out c , A , b with numbers in them. Remember the dictionary: an equality becomes *two* inequality rows, and nonnegativity gets its own rows. Report the dimensions of A .
- (3 pts)** Is the feasible set a polyhedron? Is the problem convex? One sentence each.
- (3 pts)** Orders change: only *whole* pallets and drums may fly ($x \in \mathbb{Z}^3$). Which of your answers in (c) change, and to what?

Problem 2. Convexity, by pictures**(20 points)**

No formal proofs in this problem. All you need is the *segment test* from lecture — a set is convex when the straight segment between any two of its points stays inside — plus a pencil and a ruler. Label your drawings.

- (7 pts)** Draw two convex sets in $\mathbb{R}^2$ whose *union* is not convex: mark two points of the union and the point on their segment that escapes it. Then *try* to draw two convex sets whose *intersection* is not convex. After a few honest attempts, state in one sentence what you believe to be true about intersections — no proof needed, just your pictures.
- (7 pts)** Draw the Week 1 toy $P = \{(x, y) \geq 0 : x + 3y \leq 10, 3x + y \leq 10\}$ and mark all the lattice points $X = P \cap \mathbb{Z}^2$. Now run the segment test on X : pick two points of X whose segment passes through a point that is not in X , and label that escaping point with its coordinates. What did the single line “ $\in \mathbb{Z}$ ” do to convexity?
- (6 pts)** Same picture, one question: for a max problem, why can the LP relaxation’s optimal value z_{LP} never be *smaller* than the IP’s z_{IP} ? Answer in one or two sentences, no notation needed. (Lecture’s hint: deleting the “ $\in \mathbb{Z}$ ” line only *enlarges* the set of allowed plans.)

Problem 3. Relax, round, regret**(20 points)**

A patrol packs a rucksack that carries $W = 24$ kg. Ammo boxes weigh 5 kg and are worth 8 points each; med kits weigh 7 kg and are worth 11 points each. Any nonnegative integer count of each may be packed:

$$\begin{aligned} \max \quad & 8x + 11y \\ \text{s.t.} \quad & 5x + 7y \leq 24 \\ & x, y \in \mathbb{Z}_+ \end{aligned}$$

- (a) **(5 pts)** Solve the LP relaxation graphically: draw the feasible region, list every vertex with its objective value, and report the LP optimum (x^*, y^*) and z_{LP} .
- (b) **(5 pts)** Round the LP optimum: list *every* candidate obtained by taking $\lfloor \cdot \rfloor$ or $\lceil \cdot \rceil$ in each coordinate, check each for feasibility in a table (as in lecture), and report the best *feasible* rounded value.
- (c) **(5 pts)** Find the true IP optimum by enumeration. First justify why $x \leq 4$ and $y \leq 3$ bound the search, then report the optimum and its value.
- (d) **(2 pts)** Verify the chain $z_{LP} \geq z_{IP} > \text{best rounding}$ with your numbers, and say in one sentence what rounding lost.
- (e) **(3 pts)** Implement the IP with **gurobipy** and solve it. Include a screenshot of your code and its printed output, using exactly this form:

```
print(f"optimal solution x is {x.X}")
print(f"optimal solution y is {y.X}")
print(f"optimal objective value is {model.ObjVal}")
```

Check that Gurobi agrees with your answer in (c).

Problem 4. The rewriting dictionary

(20 points)

- (a) **(10 pts)** Convert the following problem to the standard form $\max \tilde{c}^\top \tilde{x}$ s.t. $\tilde{A}\tilde{x} \leq \tilde{b}$:

$$\begin{aligned} \min \quad & 2x_1 - x_2 + 3x_3 \\ \text{s.t.} \quad & x_1 + x_2 \geq 4 \\ & 2x_1 - x_3 = 5 \\ & x_2 + x_3 \leq 6 \\ & x_1 \geq 0, \quad x_3 \geq 0, \quad x_2 \text{ free} \end{aligned}$$

List the new variable vector $\tilde{x}$, write out $\tilde{c}$, $\tilde{A}$, $\tilde{b}$ with numbers, report the dimensions of $\tilde{A}$, and state how the optimal value of the rewritten problem relates to the original one. *Follow-up:* if x_2 were declared integer, which of your new variables must be declared integer — and show by example why declaring only one of them integer is wrong.

- (b) **(10 pts)** Ten escort vehicles must be split across two convoy routes: $x_1 + x_2 = 10$, $x \geq 0$. The mission's exposure is the *worst* of three segment scores, each decreasing in the escorts assigned:

$$s_1 = 40 - 3x_1 - x_2, \quad s_2 = 38 - x_1 - 4x_2, \quad s_3 = 36 - 2x_1 - 2x_2.$$

Command wants to minimize the worst exposure: $\min_x \max\{s_1, s_2, s_3\}$.

- (i) **(5 pts)** Reformulate as an LP using the epigraph trick. Write out the full LP.
- (ii) **(2 pts)** In one sentence: why does the optimal t equal the largest s_i at the optimum, rather than sitting strictly above it?
- (iii) **(3 pts)** A teammate tries the same trick on $\min_x \min_i \{s_i(x)\}$ by writing $s_i(x) \geq t \forall i$ and maximizing t . Explain why this does not model the problem, and name the Week 2–3 tool that models “the best of several cases” correctly.

Problem 5. Switches

(20 points)

Four candidate sites for forward supply depots must together ship 100 t to the front. Opening site i costs c_i ; an open site ships $y_i \geq 0$ tons at f_i per ton, up to its capacity u_i ; a closed site ships nothing.

site i	1 (island)	2 (port)	3 (north road)	4 (south road)
fixed cost c_i (\$k)	18	12	16	14
capacity u_i (t)	80	50	70	40
per-ton cost f_i (\$k/t)	0.2	0.5	0.3	0.25

- (a) **(6 pts)** Write the base MILP: binaries x_i for opening, continuous y_i for shipping, objective, demand constraint, and one *linking constraint* per site. Why is u_i the smallest certainly-valid big- M in site i 's linking row?
- (b) **(6 pts)** Add each field rule as one or two linear constraints, and check each by trying the 0/1 cases:
- (i) site 1 is on an island supplied through the port — if site 1 opens, site 2 must open;
 - (ii) sites 3 and 4 share one mountain road — at most one of them may open;
 - (iii) command requires at least two open sites.
- (c) **(3 pts)** A new rule arrives: if *any single site* ships more than 60 t, an inspector must be hired for a flat \$5k. Model this with one new binary z and one constraint per site. Give the smallest certainly-valid M for each of the four rows — and point out which rows turn out to be vacuous, and why.
- (d) **(2 pts)** Two sentences on big- M in this model: what silently goes wrong if site 1's linking M is 50, and what goes wrong (differently) if it is 10^6 ?
- (e) **(3 pts)** Solve the full model (a)–(c) with `gurobipy` in Colab. Report which sites open, the shipments y , whether the inspector is hired, and the total cost. Then delete the inspector rule, re-solve, and report what changes.

Problem 6. Standard form, without understanding **(+20 bonus points)**

The model below is the *deterministic location–allocation problem* (LAP); it returns in Weeks 5 and 9–10. You are **not** expected to understand the problem — just apply the Week 2 dictionary.

Sites $i \in \{1, 2, 3\}$; directed arcs $a_1 = (1 \rightarrow 2)$, $a_2 = (2 \rightarrow 1)$, $a_3 = (2 \rightarrow 3)$, $a_4 = (3 \rightarrow 2)$. Decisions: open $x_i \in \{0, 1\}$, reserve $r_i \geq 0$, flow $w_a \geq 0$, shortage $q_i \geq 0$, surplus $s_i \geq 0$.

$$\begin{aligned}
 \min_{x, r, w, q, s} \quad & \sum_i c_i x_i + \sum_i f_i r_i + \sum_a t_a w_a + \pi \sum_i q_i + \sigma \sum_i s_i \\
 \text{s.t.} \quad & \sum_i r_i \leq B && \text{(budget)} \\
 & r_i \leq B x_i \quad \forall i && \text{(linking)} \\
 & \sum_i x_i \leq p && \text{(cardinality)} \\
 & \sum_{a \text{ into } i} w_a - \sum_{a \text{ out of } i} w_a + q_i - s_i + r_i = \bar{\zeta}_i \quad \forall i && \text{(balance)} \\
 & 0 \leq w_a \leq w^{\max} \quad \forall a, \quad 0 \leq q_i \leq \bar{\zeta}_i \quad \forall i, \quad r, s \geq 0, \quad x_i \in \{0, 1\} \quad \forall i
 \end{aligned}$$

site i	1	2	3	arc a	a_1	a_2	a_3	a_4
open cost c_i	100	80	120	transport t_a	4	4	6	6
reserve cost f_i	2	1	3					
mean demand $\bar{\zeta}_i$	12	8	15					

$\pi = 60$ (shortage), $\sigma = 1$ (surplus), $B = 40$, $p = 2$, $w^{\max} = 90$

Reduce everything to the course's canonical MILP form (Week 2, Section 1):

$$\max c^\top x + h^\top y \quad \text{s.t.} \quad Ax + Gy \leq b, \quad x \in \mathbb{Z}_+^3, \quad y \geq 0$$

— remember the semester convention: x = integer decisions, y = continuous.

- (a) **(4 pts)** Which decisions form the integer block x , and which the continuous block y ? Write y as a stacked vector in the order (r, w, q, s) , report the dimensions of x and y , and give the dictionary line that turns each $x_i \in \{0, 1\}$ into the form above.
- (b) **(5 pts)** Write c and h in full, with numbers. (Careful: the model is a min.)
- (c) **(9 pts)** Describe A , G , b as labeled blocks with their dimensions (budget, linking, cardinality, balance, bounds). Then write out, in full numbers — each row of A together with its row of G — these rows only: the budget row, the three linking rows, the cardinality row, and the *two* rows that encode site 2's balance equality.
- (d) **(2 pts)** Report the final dimensions of A and G , counting every row your conversion produced. (Non-negativity rows are free here — $x \in \mathbb{Z}_+^3$ and $y \geq 0$ are part of the form.)